

Mahesh Chitakoti

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SUMMARY

AI Engineer and Full Stack Developer with 2.5+ years of experience building production AI systems and scalable web applications. Specialized in RAG pipelines, agentic workflows, and vector search, with a strong backend foundation in Python, FastAPI, Node.js, and cloud-native infrastructure. Recently reduced LLM inference costs by **35%** through semantic caching on a production RAG system.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

AI & GenAI: RAG, Agentic Workflows, LangChain, OpenAI, Gemini, Vector Search, Prompt Engineering

Backend: Node.js, Express.js, NestJS, FastAPI, REST APIs, WebSockets, Microservices

Frontend: React.js, Next.js, Redux, Tailwind CSS

Databases: PostgreSQL, MongoDB, Redis, Qdrant, pgvector

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, CI/CD

EXPERIENCE

AI Engineer – SAHIONEXT Technology Pvt. Ltd.

Oct 2025 – Apr 2026

Remote

- Designed and deployed production **RAG pipelines** using **FastAPI**, **LangChain**, **Gemini APIs**, and **Redis semantic caching**, reducing LLM inference costs by **35%**.
- Developed AI microservices for transcript processing and semantic search over millions of indexed records using **Qdrant**, tuning embeddings and filters to keep query latency low at scale.
- Designed real-time streaming pipelines using **WebSockets** and **MQTT** with retry and fallback handling, significantly reducing dropped-message incidents in production.
- Partnered with product and data teams to build **React.js** + **Redux** dashboards for AI analytics and production monitoring, improving observability and speeding up incident triage.

Full Stack Developer (Contract) – PearlThoughts

Feb 2025 – Sep 2025

Remote

- Engineered backend automation systems using **NestJS**, **TypeScript**, and **Google Workspace APIs**, executing **1K+ workflows/day** across multiple client engagements.
- Developed APIs and microservices backed by **PostgreSQL** and **MongoDB**, replacing manual processing steps with automated workflows.
- Containerized services with **Docker** and deployed on **AWS** with CI/CD pipelines for reliable, repeatable releases.
- Performed QA validation for Coursera course launches, auditing coding labs and grading systems in coordination with course instructors and platform teams.

Full Stack Developer – The DevOps Team

Oct 2023 – Jan 2025

Bangalore, India

- Optimized **Node.js** + **Express.js** APIs, cutting median response time from **800ms** to **320ms** with **Redis caching** for high-frequency platform operations.
- Developed recommendation workflows for a job portal to improve candidate-job matching based on skills, preferences, and role requirements.
- Improved user onboarding conversion through A/B-tested **React.js** and **Next.js** signup and activation flows.
- Implemented real-time notifications using **WebSockets** and automated deployments with **Docker** and **Kubernetes**, collaborating with QA and DevOps teams on stable production releases.

PROJECTS

WorkBridge – AI-Powered Project Collaboration Platform

May 2026 – Present

React.js, Next.js, Node.js, Express.js, Python, PostgreSQL, Redis, OAuth | [workbridgee.vercel.app](#)

- Built and deployed an AI-powered collaboration platform that connects students with real-world industry projects through skill-based matching.
- Developed the full stack — **React.js/Next.js** interfaces for project discovery and collaboration, with **Node.js**, **Express.js**, and **PostgreSQL** services for auth, project management, and workflows.
- Currently extending the platform with AI-based recommendation workflows to match student skills to project requirements. Additional projects, including RAG and agentic AI demos, on [GitHub](#).

CERTIFICATIONS

Oracle AI Vector Search Certified Professional, by [Oracle](#) – Sep 2025

Machine Learning I, by [Columbia University](#) – Aug 2025

Oracle Cloud Infrastructure 2024 Generative AI Certified Professional, by [Oracle](#) – Aug 2024

EDUCATION

Adichunchanagiri Institute of Technology

Bachelor of Engineering, Visvesvaraya Technological University (VTU)

Chikkamagaluru, KA

Aug 2023